



Assignment: SQL Notebook for Peer Assignment

Estimated time needed: **60** minutes.

Introduction

Using this Python notebook you will:

1. Understand the SpaceX DataSet
2. Load the dataset into the corresponding table in a Db2 database
3. Execute SQL queries to answer assignment questions

Overview of the DataSet

SpaceX has gained worldwide attention for a series of historic milestones.

It is the only private company ever to return a spacecraft from low-earth orbit, which it first accomplished in December 2010. SpaceX advertises Falcon 9 rocket launches on its website with a cost of 62 million dollars whereas other providers cost upward of 165 million dollars each, much of the savings is because Space X can reuse the first stage.

Therefore if we can determine if the first stage will land, we can determine the cost of a launch.

This information can be used if an alternate company wants to bid against SpaceX for a rocket launch.

This dataset includes a record for each payload carried during a SpaceX mission into outer space.

Download the datasets

This assignment requires you to load the spacex dataset.

In many cases the dataset to be analyzed is available as a .CSV (comma separated values) file, perhaps on the internet. Click on the link below to download and save the dataset (.CSV file):

[Spacex DataSet](#)

```
In [1]: !pip install sqlalchemy==1.3.9
```

```
Collecting sqlalchemy==1.3.9
  Downloading SQLAlchemy-1.3.9.tar.gz (6.0 MB)
    ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 6.0/6.0 MB 122.4 MB/s eta 0:00:00
  Preparing metadata (setup.py) ... one
Building wheels for collected packages: sqlalchemy
  Building wheel for sqlalchemy (setup.py) ...done
  Created wheel for sqlalchemy: filename=SQLAlchemy-1.3.9-cp312-cp312-linux_x86_64.whl size=1160111 sha256=3e03d3d2532db7198ad1b424d7a50fcbbcbf94b7aa7de7fbda5bc135d2bef1ef
  Stored in directory: /home/jupyterlab/.cache/pip/wheels/b3/1c/42/0e26b8d512adc6bce10ff71a05229366b4cc6c641cd3b42111
Successfully built sqlalchemy
Installing collected packages: sqlalchemy
  Attempting uninstall: sqlalchemy
    Found existing installation: SQLAlchemy 2.0.37
    Uninstalling SQLAlchemy-2.0.37:
      Successfully uninstalled SQLAlchemy-2.0.37
ERROR: pip's dependency resolver does not currently take into account all the packages that are installed. This behaviour is the source of the following dependency conflicts.
jupyterhub 5.2.1 requires SQLAlchemy>=1.4.1, but you have sqlalchemy 1.3.9 which is incompatible.
Successfully installed sqlalchemy-1.3.9
```

Connect to the database

Let us first load the SQL extension and establish a connection with the database

```
In [2]: !pip install ipython-sql
        !pip install ipython-sql prettytable
```

```

Collecting ipython-sql
  Downloading ipython_sql-0.5.0-py3-none-any.whl.metadata (17 kB)
Collecting prettytable (from ipython-sql)
  Downloading prettytable-3.17.0-py3-none-any.whl.metadata (34 kB)
Requirement already satisfied: ipython in /opt/conda/lib/python3.12/site-packages (from ipython-sql) (8.31.0)
Collecting sqlalchemy>=2.0 (from ipython-sql)
  Downloading sqlalchemy-2.0.45-cp312-cp312-manylinux2014_x86_64.manylinux_2_17_x86_64.manylinux_2_28_x86_64.whl.metadata (9.5 kB)
Collecting sqlparse (from ipython-sql)
  Downloading sqlparse-0.5.5-py3-none-any.whl.metadata (4.7 kB)
Requirement already satisfied: six in /opt/conda/lib/python3.12/site-packages (from ipython-sql) (1.17.0)
Requirement already satisfied: ipython-genutils in /opt/conda/lib/python3.12/site-packages (from ipython-sql) (0.2.0)
Requirement already satisfied: greenlet>=1 in /opt/conda/lib/python3.12/site-packages (from sqlalchemy>=2.0->ipython-sql) (3.1.1)
Requirement already satisfied: typing-extensions>=4.6.0 in /opt/conda/lib/python3.12/site-packages (from sqlalchemy>=2.0->ipython-sql) (4.12.2)
Requirement already satisfied: decorator in /opt/conda/lib/python3.12/site-packages (from ipython->ipython-sql) (5.1.1)
Requirement already satisfied: jedi>=0.16 in /opt/conda/lib/python3.12/site-packages (from ipython->ipython-sql) (0.19.2)
Requirement already satisfied: matplotlib-inline in /opt/conda/lib/python3.12/site-packages (from ipython->ipython-sql) (0.1.7)
Requirement already satisfied: pexpect>4.3 in /opt/conda/lib/python3.12/site-packages (from ipython->ipython-sql) (4.9.0)
Requirement already satisfied: prompt_toolkit<3.1.0,>=3.0.41 in /opt/conda/lib/python3.12/site-packages (from ipython->ipython-sql) (3.0.50)
Requirement already satisfied: pygments>=2.4.0 in /opt/conda/lib/python3.12/site-packages (from ipython->ipython-sql) (2.19.1)
Requirement already satisfied: stack_data in /opt/conda/lib/python3.12/site-packages (from ipython->ipython-sql) (0.6.3)
Requirement already satisfied: traitlets>=5.13.0 in /opt/conda/lib/python3.12/site-packages (from ipython->ipython-sql) (5.14.3)
Requirement already satisfied: wcwidth in /opt/conda/lib/python3.12/site-packages (from prettytable->ipython-sql) (0.2.13)
Requirement already satisfied: parso<0.9.0,>=0.8.4 in /opt/conda/lib/python3.12/site-packages (from jedi>=0.16->ipython->ipython-sql) (0.8.4)
Requirement already satisfied: ptyprocess>=0.5 in /opt/conda/lib/python3.12/site-packages (from pexpect>4.3->ipython->ipython-sql) (0.7.0)
Requirement already satisfied: executing>=1.2.0 in /opt/conda/lib/python3.12/site-packages (from stack_data->ipython->ipython-sql) (2.1.0)
Requirement already satisfied: asttokens>=2.1.0 in /opt/conda/lib/python3.12/site-packages (from stack_data->ipython->ipython-sql) (3.0.0)
Requirement already satisfied: pure_eval in /opt/conda/lib/python3.12/site-packages (from stack_data->ipython->ipython-sql) (0.2.3)
Downloading ipython_sql-0.5.0-py3-none-any.whl (20 kB)
Downloading sqlalchemy-2.0.45-cp312-cp312-manylinux2014_x86_64.manylinux_2_17_x86_64.manylinux_2_28_x86_64.whl (3.3 MB)
3.3/3.3 MB 87.5 MB/s eta 0:00:00
Downloading prettytable-3.17.0-py3-none-any.whl (34 kB)
Downloading sqlparse-0.5.5-py3-none-any.whl (46 kB)
Installing collected packages: sqlparse, sqlalchemy, prettytable, ipython-sql
  Attempting uninstall: sqlalchemy
    Found existing installation: SQLAlchemy 1.3.9
    Uninstalling SQLAlchemy-1.3.9:
      Successfully uninstalled SQLAlchemy-1.3.9
Successfully installed ipython-sql-0.5.0 prettytable-3.17.0 sqlalchemy-2.0.45 sqlparse-0.5.5
Requirement already satisfied: ipython-sql in /opt/conda/lib/python3.12/site-packages (0.5.0)
Requirement already satisfied: prettytable in /opt/conda/lib/python3.12/site-packages (3.17.0)
Requirement already satisfied: ipython in /opt/conda/lib/python3.12/site-packages (from ipython-sql) (8.31.0)
Requirement already satisfied: sqlalchemy>=2.0 in /opt/conda/lib/python3.12/site-packages (from ipython-sql) (2.0.45)
Requirement already satisfied: sqlparse in /opt/conda/lib/python3.12/site-packages (from ipython-sql) (0.5.5)
Requirement already satisfied: six in /opt/conda/lib/python3.12/site-packages (from ipython-sql) (1.17.0)
Requirement already satisfied: ipython-genutils in /opt/conda/lib/python3.12/site-packages (from ipython-sql) (0.2.0)
Requirement already satisfied: wcwidth in /opt/conda/lib/python3.12/site-packages (from prettytable) (0.2.13)
Requirement already satisfied: greenlet>=1 in /opt/conda/lib/python3.12/site-packages (from sqlalchemy>=2.0->ipython-sql) (3.1.1)

```

Requirement already satisfied: typing-extensions>=4.6.0 in /opt/conda/lib/python3.12/site-packages (from sqlalchemy>=2.0->ipython-sql) (4.12.2)
 Requirement already satisfied: decorator in /opt/conda/lib/python3.12/site-packages (from ipython->ipython-sql) (5.1.1)
 Requirement already satisfied: jedi>=0.16 in /opt/conda/lib/python3.12/site-packages (from ipython->ipython-sql) (0.19.2)
 Requirement already satisfied: matplotlib-inline in /opt/conda/lib/python3.12/site-packages (from ipython->ipython-sql) (0.1.7)
 Requirement already satisfied: pexpect>4.3 in /opt/conda/lib/python3.12/site-packages (from ipython->ipython-sql) (4.9.0)
 Requirement already satisfied: prompt_toolkit<3.1.0,>=3.0.41 in /opt/conda/lib/python3.12/site-packages (from ipython->ipython-sql) (3.0.50)
 Requirement already satisfied: pygments>=2.4.0 in /opt/conda/lib/python3.12/site-packages (from ipython->ipython-sql) (2.19.1)
 Requirement already satisfied: stack_data in /opt/conda/lib/python3.12/site-packages (from ipython->ipython-sql) (0.6.3)
 Requirement already satisfied: traitlets>=5.13.0 in /opt/conda/lib/python3.12/site-packages (from ipython->ipython-sql) (5.14.3)
 Requirement already satisfied: parso<0.9.0,>=0.8.4 in /opt/conda/lib/python3.12/site-packages (from jedi>=0.16->ipython->ipython-sql) (0.8.4)
 Requirement already satisfied: ptyprocess>=0.5 in /opt/conda/lib/python3.12/site-packages (from pexpect>4.3->ipython->ipython-sql) (0.7.0)
 Requirement already satisfied: executing>=1.2.0 in /opt/conda/lib/python3.12/site-packages (from stack_data->ipython->ipython-sql) (2.1.0)
 Requirement already satisfied: asttokens>=2.1.0 in /opt/conda/lib/python3.12/site-packages (from stack_data->ipython->ipython-sql) (3.0.0)
 Requirement already satisfied: pure_eval in /opt/conda/lib/python3.12/site-packages (from stack_data->ipython->ipython-sql) (0.2.3)

In [3]: `%load_ext sql`

In [4]: `import csv, sqlite3
import prettytable
prettytable.DEFAULT = 'DEFAULT'

con = sqlite3.connect("my_data1.db")
cur = con.cursor()`

In [5]: `!pip install -q pandas`

In [6]: `%sql sqlite:///my_data1.db`

In [7]: `import pandas as pd
df = pd.read_csv("https://cf-courses-data.s3.us.cloud-object-storage.appdomain.cloud/IBM-DS0321
df.to_sql("SPACEXTBL", con, if_exists='replace', index=False, method="multi")`

Out[7]: 101

Note:This below code is added to remove blank rows from table

In [8]: `#DROP THE TABLE IF EXISTS

%sql DROP TABLE IF EXISTS SPACEXTABLE;

* sqlite:///my_data1.db
Done.`

Out[8]: []

In [9]: `%sql create table SPACEXTABLE as select * from SPACEXTBL where Date is not null

* sqlite:///my_data1.db
Done.`

Out[9]: []

Tasks

Now write and execute SQL queries to solve the assignment tasks.

Note: If the column names are in mixed case enclose it in double quotes For Example "Landing_Outcome"

Task 1

Display the names of the unique launch sites in the space mission

```
In [12]: # %sql SELECT Launch_Site FROM SPACEXTABLE

# Query uses DISTINCT to return only the unique values from the column
%sql SELECT DISTINCT Launch_Site FROM SPACEXTABLE;

* sqlite:///my_data1.db
Done.
```

```
Out[12]: Launch_Site
-----
CCAFS LC-40
VAFB SLC-4E
KSC LC-39A
CCAFS SLC-40
```

Task 2

Display 5 records where launch sites begin with the string 'CCA'

```
In [13]: %sql SELECT * FROM SPACEXTABLE WHERE "Launch_Site" LIKE 'CCA%' LIMIT 5;

* sqlite:///my_data1.db
Done.
```

```
Out[13]:
```

Date	Time (UTC)	Booster_Version	Launch_Site	Payload	PAYLOAD_MASS_KG_	Orbit	Customer	Missio
2010-06-04	18:45:00	F9 v1.0 B0003	CCAFS LC-40	Dragon Spacecraft Qualification Unit	0	LEO	SpaceX	
2010-12-08	15:43:00	F9 v1.0 B0004	CCAFS LC-40	Dragon demo flight C1, two CubeSats, barrel of Brouere cheese	0	LEO (ISS)	NASA (COTS) NRO	
2012-05-22	7:44:00	F9 v1.0 B0005	CCAFS LC-40	Dragon demo flight C2	525	LEO (ISS)	NASA (COTS)	
2012-10-08	0:35:00	F9 v1.0 B0006	CCAFS LC-40	SpaceX CRS-1	500	LEO (ISS)	NASA (CRS)	
2013-03-01	15:10:00	F9 v1.0 B0007	CCAFS LC-40	SpaceX CRS-2	677	LEO (ISS)	NASA (CRS)	

```
In [19]: %sql SELECT * FROM SPACEXTABLE;

* sqlite:///my_data1.db
Done.
```

Out[19]:

Date	Time (UTC)	Booster_Version	Launch_Site	Payload	PAYLOAD_MASS__KG_	Orbit	Customer
2010-06-04	18:45:00	F9 v1.0 B0003	CCAFS LC-40	Dragon Spacecraft Qualification Unit	0	LEO	SpaceX
2010-12-08	15:43:00	F9 v1.0 B0004	CCAFS LC-40	Dragon demo flight C1, two CubeSats, barrel of Brouere cheese	0	LEO (ISS)	NASA (COTS) NASA (NR)
2012-05-22	7:44:00	F9 v1.0 B0005	CCAFS LC-40	Dragon demo flight C2	525	LEO (ISS)	NASA (COTS)
2012-10-08	0:35:00	F9 v1.0 B0006	CCAFS LC-40	SpaceX CRS-1	500	LEO (ISS)	NASA (CRS)
2013-03-01	15:10:00	F9 v1.0 B0007	CCAFS LC-40	SpaceX CRS-2	677	LEO (ISS)	NASA (CRS)
2013-09-29	16:00:00	F9 v1.1 B1003	VAFB SLC-4E	CASSIOPE	500	Polar LEO	MDA
2013-12-03	22:41:00	F9 v1.1	CCAFS LC-40	SES-8	3170	GTO	SES
2014-01-06	22:06:00	F9 v1.1	CCAFS LC-40	Thaicom 6	3325	GTO	Thaicom
2014-04-18	19:25:00	F9 v1.1	CCAFS LC-40	SpaceX CRS-3	2296	LEO (ISS)	NASA (CRS)
2014-07-14	15:15:00	F9 v1.1	CCAFS LC-40	OG2 Mission 1 6 Orbcomm-OG2 satellites	1316	LEO	Orbcomm
2014-08-05	8:00:00	F9 v1.1	CCAFS LC-40	AsiaSat 8	4535	GTO	AsiaSat
2014-09-07	5:00:00	F9 v1.1 B1011	CCAFS LC-40	AsiaSat 6	4428	GTO	AsiaSat
2014-09-21	5:52:00	F9 v1.1 B1010	CCAFS LC-40	SpaceX CRS-4	2216	LEO (ISS)	NASA (CRS)
2015-01-10	9:47:00	F9 v1.1 B1012	CCAFS LC-40	SpaceX CRS-5	2395	LEO (ISS)	NASA (CRS)
2015-02-11	23:03:00	F9 v1.1 B1013	CCAFS LC-40	DSCOVR	570	HEO	U.S. Air Force NASA NOAA
2015-03-02	3:50:00	F9 v1.1 B1014	CCAFS LC-40	ABS-3A Eutelsat 115 West B	4159	GTO	ABS Eutelsat
2015-04-14	20:10:00	F9 v1.1 B1015	CCAFS LC-40	SpaceX CRS-6	1898	LEO (ISS)	NASA (CRS)
2015-04-27	23:03:00	F9 v1.1 B1016	CCAFS LC-40	Turkmen 52 / MonacoSAT	4707	GTO	Turkmenistan National Space Agency
2015-06-28	14:21:00	F9 v1.1 B1018	CCAFS LC-40	SpaceX CRS-7	1952	LEO (ISS)	NASA (CRS)
2015-12-22	1:29:00	F9 FT B1019	CCAFS LC-40	OG2 Mission 2 11 Orbcomm-OG2 satellites	2034	LEO	Orbcomm
2016-01-17	18:42:00	F9 v1.1 B1017	VAFB SLC-4E	Jason-3	553	LEO	NASA (LSP) NOAA CNES
2016-03-04	23:35:00	F9 FT B1020	CCAFS LC-40	SES-9	5271	GTO	SES

Date	Time (UTC)	Booster_Version	Launch_Site	Payload	PAYLOAD_MASS__KG_	Orbit	Customer
2016-04-08	20:43:00	F9 FT B1021.1	CCAFS LC-40	SpaceX CRS-8	3136	LEO (ISS)	NASA (CRS)
2016-05-06	5:21:00	F9 FT B1022	CCAFS LC-40	JCSAT-14	4696	GTO	SKY Perfect JSAT Group
2016-05-27	21:39:00	F9 FT B1023.1	CCAFS LC-40	Thaicom 8	3100	GTO	Thaicom
2016-06-15	14:29:00	F9 FT B1024	CCAFS LC-40	ABS-2A Eutelsat 117 West B	3600	GTO	ABS Eutelsat
2016-07-18	4:45:00	F9 FT B1025.1	CCAFS LC-40	SpaceX CRS-9	2257	LEO (ISS)	NASA (CRS)
2016-08-14	5:26:00	F9 FT B1026	CCAFS LC-40	JCSAT-16	4600	GTO	SKY Perfect JSAT Group
2017-01-14	17:54:00	F9 FT B1029.1	VAFB SLC-4E	Iridium NEXT 1	9600	Polar LEO	Iridium Communication
2017-02-19	14:39:00	F9 FT B1031.1	KSC LC-39A	SpaceX CRS-10	2490	LEO (ISS)	NASA (CRS)
2017-03-16	6:00:00	F9 FT B1030	KSC LC-39A	EchoStar 23	5600	GTO	EchoStar
2017-03-30	22:27:00	F9 FT B1021.2	KSC LC-39A	SES-10	5300	GTO	SES
2017-05-01	11:15:00	F9 FT B1032.1	KSC LC-39A	NROL-76	5300	LEO	NRO
2017-05-15	23:21:00	F9 FT B1034	KSC LC-39A	Inmarsat-5 F4	6070	GTO	Inmarsat
2017-06-03	21:07:00	F9 FT B1035.1	KSC LC-39A	SpaceX CRS-11	2708	LEO (ISS)	NASA (CRS)
2017-06-23	19:10:00	F9 FT B1029.2	KSC LC-39A	BulgariaSat-1	3669	GTO	Bulsatcom
2017-06-25	20:25:00	F9 FT B1036.1	VAFB SLC-4E	Iridium NEXT 2	9600	LEO	Iridium Communication
2017-07-05	23:38:00	F9 FT B1037	KSC LC-39A	Intelsat 35e	6761	GTO	Intelsat
2017-08-14	16:31:00	F9 B4 B1039.1	KSC LC-39A	SpaceX CRS-12	3310	LEO (ISS)	NASA (CRS)
2017-08-24	18:51:00	F9 FT B1038.1	VAFB SLC-4E	Formosat-5	475	SSO	NSPCC
2017-09-07	14:00:00	F9 B4 B1040.1	KSC LC-39A	Boeing X-37B OTV-5	4990	LEO	U.S. Air Force
2017-10-09	12:37:00	F9 B4 B1041.1	VAFB SLC-4E	Iridium NEXT 3	9600	Polar LEO	Iridium Communication
2017-10-11	22:53:00	F9 FT B1031.2	KSC LC-39A	SES-11 / EchoStar 105	5200	GTO	SES EchoStar
2017-10-30	19:34:00	F9 B4 B1042.1	KSC LC-39A	Koreasat 5A	3500	GTO	KT Corporation
2017-12-15	15:36:00	F9 FT B1035.2	CCAFS SLC-40	SpaceX CRS-13	2205	LEO (ISS)	NASA (CRS)
2017-12-23	1:27:00	F9 FT B1036.2	VAFB SLC-4E	Iridium NEXT 4	9600	Polar LEO	Iridium Communication

Date	Time (UTC)	Booster_Version	Launch_Site	Payload	PAYLOAD_MASS__KG_	Orbit	Customer
2018-01-08	1:00:00	F9 B4 B1043.1	CCAFS SLC-40	Zuma	5000	LEO	Northrop Grumman
2018-01-31	21:25:00	F9 FT B1032.2	CCAFS SLC-40	GovSat-1 / SES-16	4230	GTO	SES
2018-02-22	14:17:00	F9 FT B1038.2	VAFB SLC-4E	Paz Tintin A & B	2150	SSO	Hisdesat exactEarth Space
2018-03-06	5:33:00	F9 B4 B1044	CCAFS SLC-40	Hispasat 30W-6 PODSat	6092	GTO	Hispasat NovaWurk
2018-03-30	14:14:00	F9 B4 B1041.2	VAFB SLC-4E	Iridium NEXT 5	9600	Polar LEO	Iridium Communication
2018-04-02	20:30:00	F9 B4 B1039.2	CCAFS SLC-40	SpaceX CRS-14	2647	LEO (ISS)	NASA (CRS)
2018-04-18	22:51:00	F9 B4 B1045.1	CCAFS SLC-40	Transiting Exoplanet Survey Satellite (TESS)	362	HEO	NASA (LSP)
2018-05-11	20:14:00	F9 B5 B1046.1	KSC LC-39A	Bangabandhu-1	3600	GTO	Thales Alenia/BTR
2018-05-22	19:47:58	F9 B4 B1043.2	VAFB SLC-4E	Iridium NEXT 6 GRACE-FO 1, 2	6460	Polar LEO	Iridium Communication GFZ , NASA
2018-06-04	4:45:00	F9 B4 B1040.2	CCAFS SLC-40	SES-12	5384	GTO	SES
2018-06-29	9:42:00	F9 B4 B1045.2	CCAFS SLC-40	SpaceX CRS-15	2697	LEO (ISS)	NASA (CRS)
2018-07-22	5:50:00	F9 B5B1047.1	CCAFS SLC-40	Telstar 19V	7075	GTO	Telesat
2018-07-25	11:39:00	F9 B5B1048.1	VAFB SLC-4E	Iridium NEXT-7	9600	Polar LEO	Iridium Communication
2018-08-07	5:18:00	F9 B5 B1046.2	CCAFS SLC-40	Merah Putih	5800	GTO	Telkor Indonesia
2018-09-10	4:45:00	F9 B5B1049.1	CCAFS SLC-40	Telstar 18V / Apstar-5C	7060	GTO	Telesat
2018-10-08	2:22:00	F9 B5 B1048.2	VAFB SLC-4E	SAOCOM 1A	3000	SSO	CONA
2018-11-15	20:46:00	F9 B5 B1047.2	KSC LC-39A	Es hail 2	5300	GTO	Es hailSat
2018-12-03	18:34:05	F9 B5 B1046.3	VAFB SLC-4E	SSO-A	4000	SSO	Spaceflight Industries
2018-12-05	18:16:00	F9 B5B1050	CCAFS SLC-40	SpaceX CRS-16	2500	LEO (ISS)	NASA (CRS)
2018-12-23	13:51:00	F9 B5B1054	CCAFS SLC-40	GPS III-01	4400	MEO	USA
2019-01-11	15:31:00	F9 B5 B1049.2	VAFB SLC-4E	Iridium NEXT-8	9600	Polar LEO	Iridium Communication
2019-02-22	1:45:00	F9 B5 B1048.3	CCAFS SLC-40	Nusantara Satu, Beresheet Moon lander, S5	4850	GTO	PSN, SpaceIL, IA

Date	Time (UTC)	Booster_Version	Launch_Site	Payload	PAYLOAD_MASS__KG_	Orbit	Customer
2019-03-02	7:49:00	F9 B5B1051.1	KSC LC-39A	Crew Dragon Demo-1, SpaceX CRS-17	12055	LEO (ISS)	NASA (CCDev)
2019-05-04	6:48:00	F9 B5B1056.1	CCAFS SLC-40	SpaceX CRS-17, Starlink v0.9	2495	LEO (ISS)	NASA (CRS)
2019-05-24	2:30:00	F9 B5 B1049.3	CCAFS SLC-40	Starlink v0.9, RADARSAT Constellation	13620	LEO	SpaceX, Planet Labs
2019-06-12	14:17:00	F9 B5 B1051.2	VAFB SLC-4E	RADARSAT Constellation, SpaceX CRS-18	4200	SSO	Canadian Space Agency (CSA)
2019-07-25	22:01:00	F9 B5 B1056.2	CCAFS SLC-40	SpaceX CRS-18, AMOS-17	2268	LEO (ISS)	NASA (CRS)
2019-08-06	23:23:00	F9 B5 B1047.3	CCAFS SLC-40	AMOS-17, Starlink 1 v1.0	6500	GTO	Spacecom
2019-11-11	14:56:00	F9 B5 B1048.4	CCAFS SLC-40	Starlink 1 v1.0, SpaceX CRS-19	15600	LEO	SpaceX, Planet Labs
2019-12-05	17:29:00	F9 B5B1059.1	CCAFS SLC-40	SpaceX CRS-19, JCSat-18 / Kacific 1	2617	LEO (ISS)	NASA (CRS), Kacific
2019-12-17	0:10:00	F9 B5 B1056.3	CCAFS SLC-40	JCSat-18 / Kacific 1, Starlink 2 v1.0	6956	GTO	Sky Perfect JSAT, Kacific
2020-01-07	2:33:00	F9 B5 B1049.4	CCAFS SLC-40	Starlink 2 v1.0, Crew Dragon in-flight abort test	15600	LEO	SpaceX, Planet Labs
2020-01-19	15:30:00	F9 B5 B1046.4	KSC LC-39A	Crew Dragon in-flight abort test, Starlink 3 v1.0	12050	Sub-orbital	NASA (CTS)
2020-01-29	14:07:00	F9 B5 B1051.3	CCAFS SLC-40	Starlink 3 v1.0, Starlink 4 v1.0	15600	LEO	SpaceX, Planet Labs
2020-02-17	15:05:00	F9 B5 B1056.4	CCAFS SLC-40	Starlink 4 v1.0, SpaceX CRS-20	15600	LEO	SpaceX, Planet Labs
2020-03-07	4:50:00	F9 B5 B1059.2	CCAFS SLC-40	SpaceX CRS-20, Starlink 5 v1.0	1977	LEO (ISS)	NASA (CRS)
2020-03-18	12:16:00	F9 B5 B1048.5	KSC LC-39A	Starlink 5 v1.0, Starlink 6 v1.0	15600	LEO	SpaceX, Planet Labs
2020-04-22	19:30:00	F9 B5 B1051.4	KSC LC-39A	Starlink 6 v1.0, Crew Dragon Demo-2	15600	LEO	SpaceX, Planet Labs
2020-05-30	19:22:00	F9 B5B1058.1	KSC LC-39A	Crew Dragon Demo-2, Starlink 7 v1.0	12530	LEO (ISS)	NASA (CCDev)
2020-06-04	1:25:00	F9 B5 B1049.5	CCAFS SLC-40	Starlink 7 v1.0, Starlink 8 v1.0	15600	LEO	SpaceX, Planet Labs
2020-06-13	9:21:00	F9 B5 B1059.3	CCAFS SLC-40	Starlink 8 v1.0, SkySats-16, -17, -18, GPS III-03	15410	LEO	SpaceX, Planet Labs
2020-06-18	20:10:46	F9 B5B1060.1	CCAFS SLC-40	GPS III-03, ANASIS-II	4311	MEO	U.S. Space Force

Date	Time (UTC)	Booster_Version	Launch_Site	Payload	PAYLOAD_MASS__KG_	Orbit	Customer
2020-07-20	21:30:00	F9 B5 B1058.2	CCAFS SLC-40	ANASIS-II, Starlink 9 v1.0	5500	GTO	Republic of Korea Army Spaceflight Industries (BlackSky)
2020-08-07	5:12:00	F9 B5 B1051.5	KSC LC-39A	Starlink 9 v1.0, SXRS-1, Starlink 10 v1.0	14932	LEO	SpaceX Spaceflight Industries (BlackSky), Planet Labs
2020-08-18	14:31:00	F9 B5 B1049.6	CCAFS SLC-40	Starlink 10 v1.0, SkySat-19, -20, -21, SAOCOM 1B	15440	LEO	SpaceX, Planet Labs, Planet Labs
2020-08-30	23:18:00	F9 B5 B1059.4	CCAFS SLC-40	SAOCOM 1B, GNOMES 1, Tyvak-0172	3130	SSO	CONAE, Planet Labs, SpaceX
2020-09-03	12:46:14	F9 B5 B1060.2	KSC LC-39A	Starlink 11 v1.0, Starlink 12 v1.0	15600	LEO	SpaceX
2020-10-06	11:29:34	F9 B5 B1058.3	KSC LC-39A	Starlink 12 v1.0, Starlink 13 v1.0	15600	LEO	SpaceX
2020-10-18	12:25:57	F9 B5 B1051.6	KSC LC-39A	Starlink 13 v1.0, Starlink 14 v1.0	15600	LEO	SpaceX
2020-10-24	15:31:34	F9 B5 B1060.3	CCAFS SLC-40	Starlink 14 v1.0, GPS III-04	15600	LEO	SpaceX
2020-11-05	23:24:23	F9 B5B1062.1	CCAFS SLC-40	GPS III-04, Crew-1	4311	MEO	US Space Force
2020-11-16	0:27:00	F9 B5B1061.1	KSC LC-39A	Crew-1, Sentinel-6 Michael Freilich	12500	LEO (ISS)	NASA (CCP)
2020-11-21	17:17:08	F9 B5B1063.1	VAFB SLC-4E	Sentinel-6 Michael Freilich, Starlink 15 v1.0	1192	LEO	NASA / NOAA, ESA, EUMETSAT
2020-11-25	2:13:00	F9 B5 B1049.7	CCAFS SLC-40	Starlink 15 v1.0, SpaceX CRS-21	15600	LEO	SpaceX
2020-12-06	16:17:08	F9 B5 B1058.4	KSC LC-39A	SpaceX CRS-21	2972	LEO (ISS)	NASA (CRS)

Task 3

Display the total payload mass carried by boosters launched by NASA (CRS)

```
In [15]: %sql SELECT SUM("PAYLOAD_MASS__KG_") as Total_Payload_Mass FROM SPACEXTABLE WHERE Customer = 'NASA (CRS)'
* sqlite:///my_data1.db
Done.
Out[15]: Total_Payload_Mass
         45596
```

Task 4

Display average payload mass carried by booster version F9 v1.1

```
In [16]: %sql SELECT AVG("PAYLOAD_MASS__KG_") as Avg_payload_mass FROM SPACEXTABLE WHERE "Booster_Versio
* sqlite:///my_data1.db
Done.
```

```
Out[16]: Avg_payload_mass
2928.4
```

Task 5

List the date when the first succesful landing outcome in ground pad was acheived.

Hint: Use min function

```
In [17]: %sql SELECT MIN("Date") as First_Successful_Ground_Landing FROM SPACEXTABLE WHERE "Landing_Outc
* sqlite:///my_data1.db
Done.
```

```
Out[17]: First_Successful_Ground_Landing
2015-12-22
```

```
In [18]: %sql SELECT * FROM SPACEXTABLE WHERE "Landing_Outcome" = "Success (ground pad)";
* sqlite:///my_data1.db
Done.
```

```
Out[18]:
```

Date	Time (UTC)	Booster_Version	Launch_Site	Payload	PAYLOAD_MASS__KG_	Orbit	Customer	Mission
2015-12-22	1:29:00	F9 FT B1019	CCAFS LC-40	OG2 Mission 2 Orbcomm-OG2 satellites	2034	LEO	Orbcomm	
2016-07-18	4:45:00	F9 FT B1025.1	CCAFS LC-40	SpaceX CRS-9	2257	LEO (ISS)	NASA (CRS)	
2017-02-19	14:39:00	F9 FT B1031.1	KSC LC-39A	SpaceX CRS-10	2490	LEO (ISS)	NASA (CRS)	
2017-05-01	11:15:00	F9 FT B1032.1	KSC LC-39A	NROL-76	5300	LEO	NRO	
2017-06-03	21:07:00	F9 FT B1035.1	KSC LC-39A	SpaceX CRS-11	2708	LEO (ISS)	NASA (CRS)	
2017-08-14	16:31:00	F9 B4 B1039.1	KSC LC-39A	SpaceX CRS-12	3310	LEO (ISS)	NASA (CRS)	
2017-09-07	14:00:00	F9 B4 B1040.1	KSC LC-39A	Boeing X-37B OTV-5	4990	LEO	U.S. Air Force	
2017-12-15	15:36:00	F9 FT B1035.2	CCAFS SLC-40	SpaceX CRS-13	2205	LEO (ISS)	NASA (CRS)	
2018-01-08	1:00:00	F9 B4 B1043.1	CCAFS SLC-40	Zuma	5000	LEO	Northrop Grumman	Success statu

Task 6

List the names of the boosters which have success in drone ship and have payload mass greater than 4000 but less than 6000

```
In [22]: %sql SELECT "Booster_Version" FROM SPACEXTABLE WHERE "Landing_Outcome" = "Success (drone ship)"
* sqlite:///my_data1.db
Done.
```

```
Out[22]: Booster_Version
```

```
F9 FT B1022
```

```
F9 FT B1026
```

```
F9 FT B1021.2
```

```
F9 FT B1031.2
```

Task 7

List the total number of successful and failure mission outcomes

```
In [23]: %sql SELECT "Mission_Outcome", COUNT(*) as Total_Count FROM SPACEXTABLE GROUP BY "Mission_Outcome"
* sqlite:///my_data1.db
Done.
```

```
Out[23]: Mission_Outcome Total_Count
```

```
Failure (in flight) 1
```

```
Success 98
```

```
Success 1
```

```
Success (payload status unclear) 1
```

Task 8

List all the booster_versions that have carried the maximum payload mass, using a subquery with a suitable aggregate function.

```
In [24]: %sql SELECT "Booster_Version" FROM SPACEXTABLE WHERE "PAYLOAD_MASS_KG_" = (SELECT MAX("PAYLOAD_MASS_KG_") FROM SPACEXTABLE)
* sqlite:///my_data1.db
Done.
```

```
Out[24]: Booster_Version
```

```
F9 B5 B1048.4
```

```
F9 B5 B1049.4
```

```
F9 B5 B1051.3
```

```
F9 B5 B1056.4
```

```
F9 B5 B1048.5
```

```
F9 B5 B1051.4
```

```
F9 B5 B1049.5
```

```
F9 B5 B1060.2
```

```
F9 B5 B1058.3
```

```
F9 B5 B1051.6
```

```
F9 B5 B1060.3
```

```
F9 B5 B1049.7
```

Task 9

List the records which will display the month names, failure landing_outcomes in drone ship ,booster versions, launch_site for the months in year 2015.

Note: SQLite does not support monthnames. So you need to use substr(Date, 6,2) as month to get the months and substr(Date,0,5)='2015' for year.

```
In [25]: %sql SELECT substr(Date, 6, 2) as Month, Landing_Outcome, Booster_Version, Launch_Site FROM SPACEXTABLE WHERE substr(Date,0,5)='2015'
```

```
* sqlite:///my_data1.db
Done.
```

```
Out[25]:
```

	Month	Landing_Outcome	Booster_Version	Launch_Site
	01	Failure (drone ship)	F9 v1.1 B1012	CCAFS LC-40
	04	Failure (drone ship)	F9 v1.1 B1015	CCAFS LC-40

Task 10

Rank the count of landing outcomes (such as Failure (drone ship) or Success (ground pad)) between the date 2010-06-04 and 2017-03-20, in descending order.

```
In [26]: %sql SELECT Landing_Outcome, COUNT(*) as Count FROM SPACEXTABLE WHERE Date BETWEEN '2010-06-04'
* sqlite:///my_data1.db
Done.
```

```
Out[26]:
```

	Landing_Outcome	Count
	No attempt	10
	Success (drone ship)	5
	Failure (drone ship)	5
	Success (ground pad)	3
	Controlled (ocean)	3
	Uncontrolled (ocean)	2
	Failure (parachute)	2
	Precluded (drone ship)	1

Reference Links

- [Hands-on Lab : String Patterns, Sorting and Grouping](#)
- [Hands-on Lab: Built-in functions](#)
- [Hands-on Lab : Sub-queries and Nested SELECT Statements](#)
- [Hands-on Tutorial: Accessing Databases with SQL magic](#)
- [Hands-on Lab: Analyzing a real World Data Set](#)

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